Engine Model(s): PT6A

70-00-00

STANDARD PRACTICES - REMOVAL/INSTALLATION

1. General

A. To ensure proper reinstallation, tag and mark all parts, clips, and brackets as to their location.

B. During removal of tubes or engine parts look for indications of scoring, burning or other undesirable conditions. To facilitate reinstallation, observe the location of each part during removal. Tag unserviceable parts and units for investigation and possible repair.

C. Extreme care shall be taken to prevent dust, dirt, lockwire, nuts, washers or other foreign matter from entering the engine. It cannot be overemphasized that this precaution applies whenever work is done on the engine either on the wing or off the wing. Suitable plugs, caps, and other covering shall be used to protect all openings as they are exposed.

CAUTION: DUST CAPS USED TO PROTECT OPEN TUBES AGAINST CONTAMINATION SHALL ALWAYS BE INSTALLED OVER THE TUBE ENDS AND NOT IN THE TUBE ENDS. FLOW THROUGH THE LINES MAY BE BLOCKED OFF IF LINES ARE INADVERTENTLY INSTALLED WITH DUST CAPS IN THE TUBE ENDS.

D. If at any time items are dropped into the engine, the assembly process must stop until the dropped articles are located, even though this may require a considerable amount of time and labor. Before assembling or installing any part, be sure it is thoroughly clean.

E. Lockwire, lockwashers, tablocks, tabwashers, or cotterpins must never be reused. All lockwire and cotterpins must fit snugly in holes drilled in studs and bolts for locking purposes. Install a cotterpin so that the head fits into the castellation of the nut, and unless otherwise specified, bend one end of the pin back over the stud or bolt and the other end down flat against the nut. Only lockwire and cotterpins made of corrosion resistant steel shall be used. Bushing plugs shall be lockwired to the assembly boss or case. Do not lockwire the plug to the bushing.

F. Replace all gaskets, packings, and rubber parts at reassembly. Make sure that new nonmetallic parts to be installed (such as an oil seal) show no sign of having deteriorated in storage.

CAUTION: TOOL SURFACES WHICH MAY CONTACT ENGINE CRITICAL AREAS MUST BE ADEQUATELY COVERED WITH A PROTECTIVE MATERIAL.

G. To protect critical areas of engine parts (such as compressor and turbine disks) against scratches and nicks, tool surfaces contacting these areas must be covered with protective material.

H. During replacement of components, note condition of any removed preformed packings. Pieces accidentally torn or cut from packings must be retrieved, regardless of size, even if this entails a systematic disassembly and cleaning of the system. Ensure that new packings are free of cuts, flashings and deformities which may be sheared off at installation.

I. When installing engine parts that require the use of a hammer to facilitate assembly or installation, use only a plastic or rawhide hammer.

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J. Whenever adhesive tape has been applied to any part, the tape must be removed and the part thoroughly cleaned of all tape residue with petroleum solvent prior to being subjected to high temperature during engine run. Test results indicate that all tapes are capable of causing surface attack and/or reduction in tensile ductility as temperature increases. Do not leave tape or tape residue on engine parts.

K. If any part has been coated with corrosion preventive compounds, all traces of this compound and accumulated foreign matter must be removed.

2. Consumable Materials

The consumable materials listed below are used in the following procedures.

Item No.	Name
PWC03-001	Oil, Engine Lubricating
PWC06-002	Petrolatum, White

3. Special Tools

Not Applicable

4. Fixtures, Equipment and Supplier Tools

Not Applicable

5. Removal/Installation

CAUTION: DO NOT BEND, TWIST OR FORCE HYDRAULIC AND PNEUMATIC TUBES, WHETHER RIGID OR FLEXIBLE, TO ACHIEVE ALIGNMENT WITH MATING FITTING. ADJUST FITTING TO ALIGN WITH TUBE. CLAMP AND SECURE TUBE ONLY WHEN IT IS INSTALLED IN POSITION WITH NO NOTICEABLE TENSION OR LOAD.

A. Removal of Straight Nipples, Straight Adapters, Bulkhead Couplings and Tube Connector Nipples

(1) Remove fittings from engine lines or accessory units.

(2) Remove preformed packings from fittings and discard.

B. Removal of Tube-to-Boss Elbows, Elbow Adapters and Elbow Assemblies

(1) Remove fittings from engine lines or accessory units.

NOTE: Before removal, mark angular position of fitting on engine or unit.

(2) Remove preformed packings, packing retainers (back-up rings) and discard. Discard jam nuts from fittings.

C. Installation of Straight Nipples, Straight Adapters, Bulkhead Couplings and Tube Connector Nipples

(1) Lubricate new preformed packing with light film of engine oil (PWC03-001) or fluid to be used in line.

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(2) Install preformed packing on fitting and screw fitting in boss or flange, as applicable.

CAUTION: EXCESSIVE TIGHTENING WILL RESULT IN DAMAGE TO THREADS OF MATING PARTS.

(3) Tighten fitting in boss and torque to value detailed in relevant assembly instruction.

(4) Install jam nut on flange mounted fittings and torque jam nut to recommended value (Ref. Table 401).

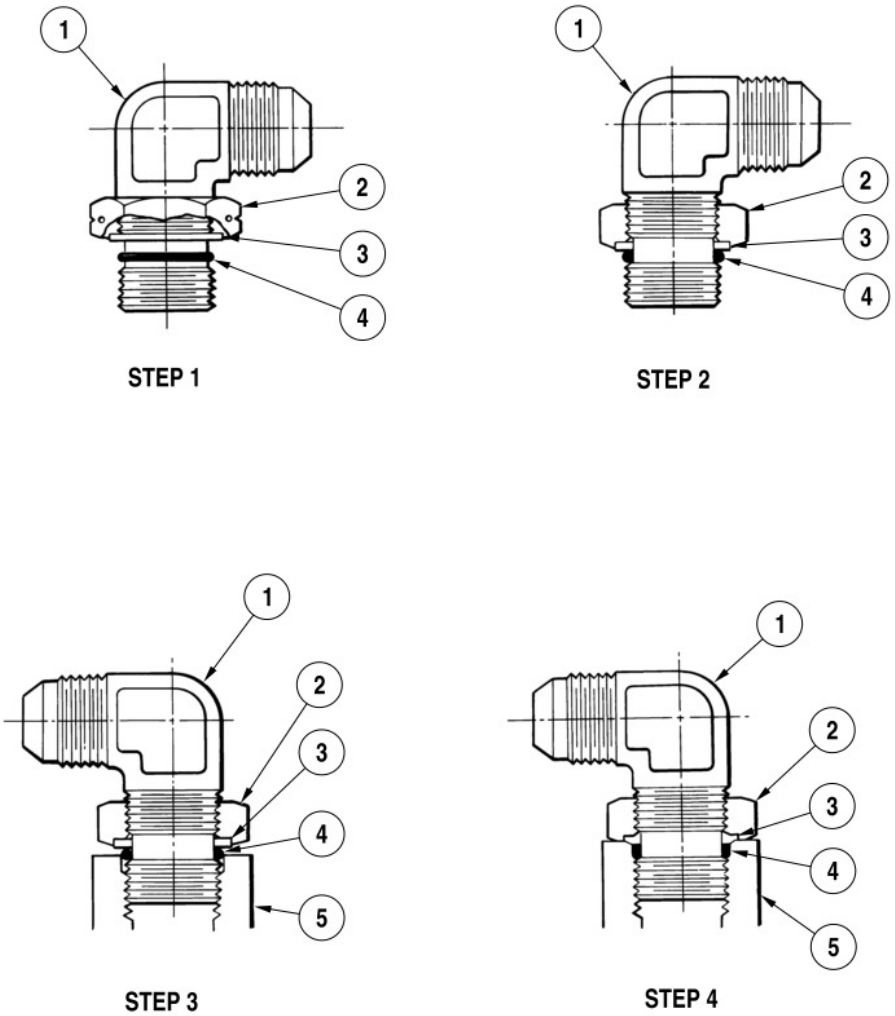
Table 401 Torque Values for Jam Nuts

THREAD SIZE	TORQUE (lb.in.)
0.250 -28	14 to 16
0.3125-24	22 to 24
0.375 -24	28 to 32
0.4375-20	38 to 42
0.500 -20	58 to 62
0.5625-18	70 to 80
0.625 -18	95 to 105
0.750 -16	145 to 155
0.8125-16	165 to 185

D. Installation of Tube-to-Boss Elbows, Elbow Adapters and Elbow Assemblies (Ref. Fig. 401)

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Figure 401 Installation of Typical Elbow Fitting



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1. Elbow Fitting
2. Jam Nut
3. Back-up Ring
4. Preformed Packing
5. Boss

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(1) Lubricate new preformed packings, packing retainer (back-up ring) and thread of elbow fitting with light film of petrolatum (PWC06-002) or fluid to be used in the line.

(2) Assemble jam nut, new packing retainer and preformed packing on elbow fitting, pressing packing retainer into counterbore of jam nut.

(3) Turn jam nut down elbow fitting until preformed packing is seated in non-threaded annulus of fitting.

(4) Turn jam nut outward until preformed packing is pushed firmly against first lower thread of fitting.

(5) Install elbow fitting into boss on unit, allowing jam nut to turn with fitting until preformed packing contacts boss mating face. This point will be recognized by increase in torque.

(6) With fitting in this position, hold jam nut stationary and turn elbow fitting into boss a further 1 1/2 turns.

NOTE: From this position the elbow fitting may be further turned inward to a maximum of one turn to facilitate alignment of fitting (Ref. Para. 1.B). Should the fitting tighten in the jam nut before completion of initial 1 1/2 turns or during final alignment, the jam nut may be allowed to turn with the fitting for the remainder of the distance.

(7) With elbow fitting in correct alignment position, tighten jam nut and torque to recommended torque (Ref. Table 401).

NOTE: Metal-to-metal contact between jam nut and boss must be obtained without exceeding recommended torque and there must be no extrusion of preformed packing or packing retainer.

E. Installation/Removal of Hydraulic, Oil and Pneumatic Lines

CAUTION: DO NOT USE FORCE TO POSITION LINE ON TO COUPLING ADAPTERS.

NOTE: Refer to specific chapter for assembly instructions; Chapter 73 for fuel and pneumatic lines, Chapter 79 for oil lines.

(1) When installing hydraulic, oil or pneumatic lines, whether solid tubes or flexible hoses, ensure lines engage on to respective coupling adapters without tension or load.

(2) Alignment must be obtained by adjusting the position of the adapters. Refer to paragraphs C and D preceeding, for instructions pertaining to the installation of adapters.

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